Marked deformations and long fillings of Conway closures

Abstract

We deform the complete pyramid-and-prism construction of a completed-root Conway closure. A holomorphic potential defined directly from the cells gives the actual meridian and disk-longitude logarithms on every crossing circle. Entire complex face maps retain the original closed cut, all group relations and every parabolic main cusp. The primitive equations have a unique small solution for independently sufficiently long signed fillings. We verify that the logarithmic meridian winds once, so the completion is the original smooth filled exterior. The absolute volume is the real part of a corrected critical action, with its constant fixed by the original complete cell volumes. The proof is classical and does not use a quantum packet or a noncancellation theorem.

1 The marked exteriors and the filling theorem

The starting link is specified by a finite ordered construction. A two-string box has endpoints $0, 1, 2, 3$ in cyclic order, starting at the bottom left. In a vertical stack FG, the lower box F is joined to the upper box G by F_2/G_1 and F_3/G_0. The exposed tuple is (F_0, F_1, G_2, G_3). A clockwise turn rotates the entire box and all its crossing disks; the new tuple is (F_1, F_2, F_3, F_0). Cylindrical closure joins 0 to 3 and 1 to 2.

Let h be the quarter-turn of an elementary crossing circle surrounding two untwisted strands. The completed grammar is

$$E ::= hh \mid \mathsf{T}(E) \mid E_1E_2 \mid hE \mid Eh, \qquad \text{closure } C(FG), \quad F, G \text{ completed.} \tag{1.1}$$

Every leaf is a distinct crossing circle. Both closing children must be completed; a bare closing leaf is not included. Fix a retained half twist $\varepsilon_\ell \in \{-1, 0, 1\}$ at each leaf. The original augmented exterior is denoted Y_E^ε. Its crossing-circle meridian μ_ℓ and disk longitude λ_ℓ are fixed before any deformation. They are the primitive curves of [1, Section 7].

For integers p_ℓ, fill the auxiliary tori along

$$\mu_\ell + p_\ell\lambda_\ell. \tag{1.2}$$

The physical crossing power in the original disk frame is $n_\ell = \varepsilon_\ell - 2p_\ell$. Indeed, cut along the twice-punctured disk and rotate one side through p_ℓ full turns before removing its auxiliary circle. The slope (1.2) becomes the meridian of the inserted solid torus, and the strands acquire $-2p_\ell$ crossings. These operations take place in the disjoint original leaf balls. They preserve the ambient S^3 and give precisely the diagram with these crossing powers. Write $Y_E^\varepsilon(\mathbf{p})$ for its exterior. If the main endpoint matching is one component, it is $S^3 \setminus K_{\varepsilon,\mathbf{p}}$. Such half-twist choices exist by [1, Proposition 8.1]; full twists preserve that component count.

We use two precise complete inputs. The strict stationary point and its negative definite internal-wire Hessian are [2, Proposition 5.3], for the grammar (1.1). The actual complete

metric, its absolute volume $2\pi F_0$, the labeled original cut and the primitive cusp basis are [1, Theorem 1.2 and Section 7]. At leaf ℓ its complete angle a_ℓ satisfies

$$\frac{1}{4} < a_\ell < \frac{\arctan 2}{\pi}, \qquad \tau_\ell = \frac{2\mathrm{i}c_\ell}{1-\varepsilon_\ell \mathrm{i}c_\ell}, \quad c_\ell = \cot(\pi a_\ell), \quad \operatorname{Im}\tau_\ell > 0. \tag{1.3}$$

Here τ_ℓ is the longitude translation divided by the meridian translation, with the orientation of the original torus. No filled metric is an input.

Section 2 defines a holomorphic classical potential $\Phi_\varepsilon(\boldsymbol{\chi})$ from these cells, with $\Phi_\varepsilon(0) = F_0$. Its variables are centered auxiliary meridian coordinates, not the meridian of the main knot.

Theorem 1.1. *For the fixed construction (1.1) and fixed half-twist pattern, there is a holomorphic framed representation $\rho_{\boldsymbol{\chi}}$ of the original group $\pi_1(Y_E^\varepsilon)$ near $\boldsymbol{\chi} = 0$. It continues the oriented complete holonomy and has primitive similarity logarithms*

$$U_\ell = -2\pi\mathrm{i}\chi_\ell, \qquad V_\ell = -2\,\partial_{\chi_\ell}\Phi_\varepsilon. \tag{1.4}$$

Every main peripheral group remains parabolic. There is $B_E > 0$ such that for every signed integer tuple with $|p_\ell| > B_E$ the function

$$S_{\varepsilon,\mathbf{p}}(\boldsymbol{\chi}) = \Phi_\varepsilon(\boldsymbol{\chi}) + \frac{\pi\mathrm{i}}{2}\sum_\ell \frac{(\chi_\ell - 1)^2}{p_\ell} \tag{1.5}$$

has a unique critical point in a fixed small neighborhood of zero. Its representation completes to a smooth oriented complete finite-volume hyperbolic metric on the original $Y_E^\varepsilon(\mathbf{p})$, and

$$\operatorname{Vol}(Y_E^\varepsilon(\mathbf{p})) = 2\pi \operatorname{Re} S_{\varepsilon,\mathbf{p}}(\boldsymbol{\chi}_*). \tag{1.6}$$

The bounds are independent of relative signs and relative sizes of the p_ℓ. Every main cusp is left complete.

Uniqueness here is local uniqueness in this prescribed family and coordinate neighborhood. It does not classify the whole character variety or all possible representations of the filled group. The action in the theorem is defined geometrically below. Identifying it with a particular finite quantum source is a separate interface, described in Section 8.

We import the general near-complete metric implication of [3, Proposition 6.1], after verifying every hypothesis on this original family. We also use its local peripheral-volume differential [3, equation (7.3)] and the same-chain comparison of [3, Section 7.3]. Neither import assumes a source asymptotic. The normalizations and applicability are made explicit in Sections 6–7.

2 A classical holomorphic cell potential

2.1 Logarithms, hidden leaf parameters and the half twist

On a complex neighborhood of each strict real angle interval continue the branch

$$J(u) = \int_0^u \operatorname{Log}(2\sin\pi s)\,\mathrm{d}s$$

from the real log-sine primitive. All occurrences are continued from the fixed complete point, so their constants are fixed. The product function is

$$\begin{aligned} m(x,y,z) = {} & J(x+y) + J(z) - J(x) - J(y) \\ & - J(z-x) - J(z-y) - J(x+y-z). \end{aligned} \tag{2.1}$$

Its strict complete domain is $\max(x,y) < z < x+y < 1$, with $x, y > 0$. For a leaf define

$$g(x,t) = J(2x) - 2J(x) - J(x-t) - J(x+t), \qquad v = g_t = \operatorname{Log} \frac{\sin \pi(x-t)}{\sin \pi(x+t)}. \tag{2.2}$$

The parameter t will be an auxiliary coordinate before the half-twist shear. At $t = 0$ this is the complete leaf action, and $v = 0$.

For $\varepsilon = 0, 1, -1$ define $t = t(x, \chi)$ by

$$\chi = t - \frac{\varepsilon v(x,t)}{2\pi \mathrm{i}}. \tag{2.3}$$

At $(x,t) = (a,0)$ its t derivative is $1 - \varepsilon \mathrm{i} \cot \pi a \neq 0$. The holomorphic implicit-function theorem gives a unique small $t(x,\chi)$, with $t(x,0) = 0$. Set

$$f_\varepsilon(x,\chi) = g(x,t) - \frac{\varepsilon v(x,t)^2}{4\pi \mathrm{i}}. \tag{2.4}$$

The elementary differential calculation

$$\mathrm{d} f_\varepsilon = g_x \, \mathrm{d}x + v \, \mathrm{d}\chi \tag{2.5}$$

is exact. The term subtracted in (2.4) keeps the same boundary covector when the meridian is sheared. It is fixed by the primitive collar words in Section 4, rather than by a choice of asymptotic phase.

2.2 Eliminating the internal wires

Let $\mathbf{y}$ be any independent coordinates for the internal wires of the completed closure after its final forced gate has been removed. The root relation is retained exactly: its outputs are $b, 1-b$. Turns likewise replace a wire by its complement. On the same affine constraint space define

$$\mathcal{F}_\varepsilon(\mathbf{y}, \boldsymbol{\chi}) = \sum_\ell f_{\varepsilon_\ell}(x_\ell(\mathbf{y}), \chi_\ell) + \sum_{\text{products}} m(x,y,z). \tag{2.6}$$

At $\boldsymbol{\chi} = 0$ this is precisely the complete real action. Its internal Hessian at the unique strict point $\mathbf{y}_0$ is nonsingular by the stated Newton input. The holomorphic implicit-function theorem therefore gives a unique small solution

$$\partial_{\mathbf{y}} \mathcal{F}_\varepsilon(\mathbf{y}_*(\boldsymbol{\chi}), \boldsymbol{\chi}) = 0, \qquad \mathbf{y}_*(0) = \mathbf{y}_0. \tag{2.7}$$

Define the classical stationary potential by

$$\Phi_\varepsilon(\boldsymbol{\chi}) = \mathcal{F}_\varepsilon(\mathbf{y}_*(\boldsymbol{\chi}), \boldsymbol{\chi}). \tag{2.8}$$

The envelope rule and (2.5) give $\partial_{\chi_\ell} \Phi_\varepsilon = v_\ell$. Since g is even in t and v is odd, (2.3) is odd in t and f_ε is even in χ. Uniqueness in (2.7) makes the stationary wires and Φ_ε separately even in each χ_ℓ.

Lemma 2.1. *The complete auxiliary Hessian is*

$$\partial_{\chi_i} \partial_{\chi_j} \Phi_\varepsilon(0) = \pi \mathrm{i} \tau_i \, \delta_{ij}. \tag{2.9}$$

Proof. At one leaf, with $c = \cot \pi a$, $g_{tt}(a,0) = -2\pi c$ and $\chi_t(a,0) = 1 - \varepsilon \mathrm{i} c$. Thus $f_{\chi\chi}(a,0) = -2\pi c/(1 - \varepsilon \mathrm{i} c) = \pi \mathrm{i} \tau$. Also $g_{xt}(x,0) = 0$, so every mixed internal-wire/auxiliary derivative vanishes at zero. Stationary-wire elimination therefore adds no Schur-complement correction to this auxiliary block. Distinct leaves have no direct mixed term in (2.6). This proves the diagonal formula. □

This argument used the cell functions and the complete variational Hessian only. A finite-level interpolation or a nonzero quantum coefficient is not needed to construct (2.8).

3 The deformed cells and the original group

3.1 A mixed leaf and its two projection charts

For real x, t near $a, 0$, put $E = e^{2\pi i x}$ and $T = e^{2\pi i t}$. The mixed leaf has apex ∞ and ordered quadrilateral

$$(C_0, C_1, C_2, C_3) = (1, E/T, -1/T, -E). \tag{3.1}$$

The successive half-arcs of its boundary circle are $x - t$, $1/2 - x$, $x + t$, $1/2 - x$, all positive nearby. Triangulation across C_0C_2 gives the triples $(x - t, 1/2 - x, 1/2 + t)$ and $(1/2 - t, x + t, 1/2 - x)$. Their complementary intermediate Lobachevsky terms cancel. Using the normalized tetrahedron formula of [1, equation (2.1)] gives exactly $\pi g(x, t)$. This fixes the classical local constant.

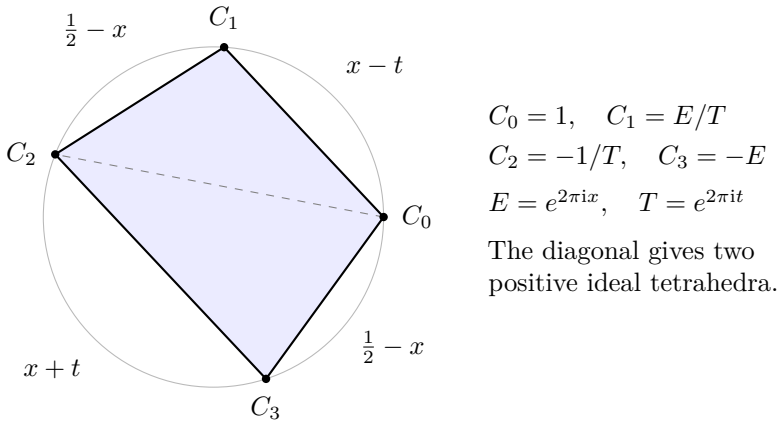


Figure 1: The boundary-plane coordinates of the mixed pyramid for $x = 0.30$, $t = 0.06$. The apex is ∞. The four successive half-arcs shown here are the actual parameters of (3.1). At $t = 0$ the quadrilateral becomes the rectangle of the complete leaf.

Its ordered cross ratio and normalized fourth vertex are

$$\begin{aligned} \varkappa &= \frac{\sin \pi(x - t) \sin \pi(x + t)}{\cos^2 \pi x} = e^{-g_x}, \\ R &= \frac{1}{1 + \varkappa} = \frac{\cos^2 \pi x}{\cos^2 \pi t} = \frac{T(1 + E)^2}{E(1 + T)^2}. \end{aligned} \tag{3.2}$$

Normalize its ordered base to $(0, \infty, 1, R)$. The plus-t apex is

$$Z = \frac{1 + E}{1 + T}, \qquad Z^2 - (1 + E)Z + ER = 0. \tag{3.3}$$

The other root is TZ, the apex of the minus-t leaf in this same normalization. The second projection chart is the reflected minus-t chart, whose apex is Z/E. It is not the reflection of the plus-t apex when $t \neq 0$.

These formulas now define holomorphic projective coordinates for complex x, t near the complete point. Reflection of coefficients is implemented by replacing the phase variables by their reciprocals. Thus the reflected minus apex is algebraically Z/E, without antiholomorphic dependence on a parameter. Every defining face-map triple remains pairwise distinct after shrinking the neighborhood; the two auxiliary fixed points themselves coalesce at completeness.

3.2 Whole complex face matching

In the normalized chart the four white or completion maps are

$$\begin{aligned} W_0 &= \begin{pmatrix} 1 & 0 \\ 0 & E \end{pmatrix}, & W_1 &= \begin{pmatrix} 1 & E-1 \\ 0 & E \end{pmatrix}, \\ W_2 &= \begin{pmatrix} E-R & R(1-E) \\ E-1 & 1-ER \end{pmatrix}, & W_3 &= \begin{pmatrix} R & 0 \\ 1-E & ER \end{pmatrix}. \end{aligned} \tag{3.4}$$

The pure shaded map and the two odd cross-copy maps are

$$S = \begin{pmatrix} 0 & -ER \\ 1 & -1-E \end{pmatrix}, \quad A = \begin{pmatrix} R & -R \\ 1 & ER-1-E \end{pmatrix}, \quad B = \begin{pmatrix} R/E-1-1/E & R \\ -1 & R \end{pmatrix}. \tag{3.5}$$

All matrices are projective. For a mixed leaf, W_0, W_2 fix their two main corners and send Z to Z/E. The apex equation (3.3) verifies this last statement also for W_2. The shaded assignments are

$$S : (Z, \infty, 1) \mapsto (Z, 0, R), \quad A : (Z, \infty, 1) \mapsto (Z/E, R, 0), \quad B : (Z, R, 0) \mapsto (Z/E, \infty, 1).$$

Each is the map of the original labeled triangle. The expressions W_1, W_3 are not extra leaf faces: their actual white pieces come from its adjacent prism.

To check those pieces throughout the expression, keep the original prism formula with complex wire parameters. Write

$$\alpha = e^{2\pi \mathrm{i}(z-x)}, \quad \beta = e^{2\pi \mathrm{i}(x+y-z)}, \quad \gamma = e^{2\pi \mathrm{i}(z-y)}.$$

Its world coordinates are

$$\begin{aligned} &v_0 = 1, \quad v_1 = \alpha, \quad v_2 = \alpha\beta, \quad v_3 = \alpha\beta\gamma, \\ &P = 1 + \alpha - \frac{\alpha[\alpha\beta(1+\gamma) - 1 - \alpha]}{\alpha^2\beta^2\gamma - \alpha}. \end{aligned} \tag{3.6}$$

The lower, upper and output ordered quadrilaterals are (v_0, v_1, P, ∞), (∞, P, v_2, v_3) and (v_0, v_1, v_2, v_3), with wire phases $\beta\gamma, \alpha\beta, \alpha\beta\gamma$. For an ordered tuple (a, b, c, d) let

$$M_{a,b,c}(z) = \frac{(z-a)(c-b)}{(z-b)(c-a)}, \qquad R = M_{a,b,c}(d).$$

Its world white maps are $M^{\#\,-1} W_j M$, where $\#$ reciprocates α, β, γ. At a real strict prism this agrees with reflection; it is a holomorphic rational prescription in general.

The exact rational identities

$$\begin{aligned} W_{F,1} = W_{G,1} = W_{o,1} = W_{\mathrm{top}}, \qquad W_{F,3} = W_{G,3} = W_{o,3} = W_{\mathrm{bottom}}, \\ W_{F,2} = W_{G,0}, \quad W_{o,0} = W_{F,0}, \qquad W_{o,2} = W_{G,2} \end{aligned} \tag{3.7}$$

are [1, equation (5.5)]. Their proof is a rational substitution in three independent phase variables, so it already proves (3.7) off the real chamber. Here the top and bottom maps carry the original white triangles (P, v_1, v_2) and (∞, v_0, v_3) to their reflected charts. No equality merely of absolute cross ratios is being used.

At an internal wire, (2.5) says that the child output derivative is g_x or the corresponding stationary product derivative. The internal critical equations therefore match the full complex cross ratios under the same original vertex assignments as at completeness. Three matched vertices fix a Möbius map, and the cross-ratio equation matches the fourth. The former

quadrilateral need not remain planar; the argument matches its four flags and both triangles of any fixed subdivision.

Turns and closure use $T_Q(z) = (R-1)/(z-1)$ and $C_Q(z) = (z-R)/(z-1)$, with identities

$$\begin{aligned} T_Q W_{j+1}(E,R) &= W_j(E^{-1}, 1-R) T_Q, \\ C_Q W_j(E^{-1}, R) &= W_{2-j}(E,R) C_Q. \end{aligned} \tag{3.8}$$

These are [1, equation (5.6)], again rational identities. Induction through (3.7) and (3.8) proves that every triangular piece of a given original white face has the same entire map. In particular the parent prism supplies W_1, W_3 at every elementary cap edge. The two completed root children ensure that the closing seams are white. This is the specific place where the grammar (1.1) is retained.

3.3 Relations and positivity on the retained chain

Let $S^{\#} = S(E^{-1}, R)$. Direct multiplication gives

$$\begin{aligned} W_0 S = S^{\#} W_0, \quad & W_2 S = S^{\#} W_2, \quad & W_3 S = S^{\#} W_1, \\ W_0^{-1} B W_2^{-1} A = I, \quad & W_2^{-1} B W_0^{-1} A = I, \quad & W_1^{-1} B W_3^{-1} A = I. \end{aligned} \tag{3.9}$$

The first row is the pure edge return; the second is the odd return, including its translational part. These are identities for independent E, R, as proved in [1, equations (5.7)–(5.8)]. The original cut balls have a dual two-skeleton with generators the paired faces and relations precisely these edge cycles and the white seams. Since (3.7) identifies the whole white maps, all those relations hold. The family is consequently a representation of the original $\pi_1(Y_E^{\varepsilon})$. The deformed image of a half ball is not assumed to be an embedded convex polyhedron.

For later use, retain a compatible triangulation of the complete cells. First perform the actual assembly to the single complete polyhedron P_E. Its finitely many ideal vertices are distinct by the incidence proof in [1, Section 3]. Choose an order on these vertices, and restrict it to every constituent cell and its faces. The pulling triangulations agree on internal quadrilaterals, since these are now the same labeled face in P_E, regardless of the earlier turn or attachment order. Coning from the first vertex to faces not containing it gives nondegenerate tetrahedra. The vertex order chooses their diagonals; each tetrahedron is oriented by the ambient oriented cell, rather than by the increasing global vertex order. In that orientation each complete shape is positive. Use the identical subdivision in the second labeled copy. All nontriangular external face pairings are the white maps between corresponding copies, so their subdivisions match. The pure and odd shaded pairings may permute vertices, but their faces are already triangles and contain no diagonal to choose. Thus no order on quotient cusp vertices, which could identify many different cut vertices, is assumed. Transport these finitely many vertices by the formulas above. Their complete tetrahedra are positive, so their shapes remain in the positive half-plane in one common small parameter neighborhood. The full face maps match their triangles and edge relations. We have the same oriented ideal fundamental chain and the same peripheral marking as at completeness. This construction does not require an ideal triangulation of the subsequently filled manifold.

4 Primitive logarithms and the unfilled cusps

For a peripheral element fixing a selected flag, its similarity multiplier means the coefficient of its affine action after that flag is sent to ∞. At a finite flag Z it is the reciprocal of the derivative at Z. These are projective similarity multipliers, twice the usual selected eigenvalue logarithms. All logarithms below are their small branches, zero at completeness.

The original disk and meridian words are the ones already traced at completeness:

$$\lambda = W_2^{-1}W_0, \qquad \mu_0 = S^{-1}, \qquad \mu_- = A^{-1}W_0, \qquad \mu_+ = W_0^{-1}B. \tag{4.1}$$

They remain the same based curves in the original cut. Their primitiveness and orientations are [1, equations (7.1)–(7.5)]; in particular it is the actual disk longitude, not a multiple read from an eigenvalue. Their selected common flag is Z. Put

$$\rho = \frac{Z}{1+E-Z} = \frac{1}{T}, \qquad Q = \frac{E-Z}{Z-1} = \frac{ET-1}{E-T}. \tag{4.2}$$

Inverting derivatives at Z in (4.1) gives the exact multipliers

$$\lambda : Q^2, \qquad \mu_0 : \rho, \qquad \mu_- : \rho Q, \qquad \mu_+ : \rho/Q. \tag{4.3}$$

For example the fixed-point equation of S is precisely (3.3), and substitution in the Möbius derivative $\det S/(cZ+d)^2$ gives the first meridian factor. The same substitution gives the other three factors. At $t = 0$ both ρ and Q equal one, so no central logarithmic constant can enter their small branches.

From (2.2) and (4.2) one has $Q = e^{-v}$ and $\rho = e^{-2\pi \mathrm{i} t}$. It follows that the original signed meridian and disk longitude have logarithms

$$U = -2\pi \mathrm{i} t + \varepsilon v = -2\pi \mathrm{i}\chi, \qquad V = -2v. \tag{4.4}$$

Using the envelope rule proves (1.4). The half-twist part of $G = 2\pi \mathrm{i}\Phi_\varepsilon$ is $-\varepsilon V^2/8$ at each leaf, exactly the shear in (2.4).

All main cusps also retain their actual peripheral words. At corner C_j the small main meridian is $W_{j-1}^{-1}W_j$, or its inverse according to orientation. Both factors fix that corner. Their derivatives are equal: E^{-1} at C_0, C_2 and E at C_1, C_3, using the finite coordinate at infinity. Thus the meridian multiplier is identically one. It is nonidentity at the complete point, and remains so nearby; hence it is parabolic. Every element commuting with a nonidentity parabolic Möbius map is a translation in its fixed-flag chart. The original main longitude commutes with this meridian. Its multiplier is therefore also one. This verifies the entire unfilled main peripheral group, not just a trace equation at a single corner.

The selected auxiliary flags, the main flags and all face maps vary holomorphically and stay in bounded frames near their complete values. A finite collection of charts suffices. We may therefore choose one neighborhood where the actual framed family and all the positive tetrahedra constructed above are simultaneously defined.

5 The primitive solution for independently signed slopes

Put $r_\ell = 1/p_\ell$ and solve

$$V_\ell(\boldsymbol{\chi}) + r_\ell U_\ell(\boldsymbol{\chi}) = -2\pi \mathrm{i} r_\ell. \tag{5.1}$$

At $\mathbf{r} = 0$ the solution is $\boldsymbol{\chi} = 0$. Lemma 2.1 and (1.4) give

$$D_{\boldsymbol{\chi}} V(0) = -2\pi \mathrm{i}\ \mathrm{diag}(\tau_1, \dots, \tau_H). \tag{5.2}$$

It is invertible because $\operatorname{Im}\tau_\ell > 0$. The holomorphic implicit-function theorem gives a unique solution $\boldsymbol{\chi}_*(\mathbf{r})$ on one fixed polydisk in all the independent reciprocal slopes. Shrink that polydisk so that its image lies in the framed neighborhood of Section 4. For real signed integer p_ℓ this requires only a lower bound on each $|p_\ell|$. There is no constraint on their ratios.

When $p_\ell \neq 0$, (5.1) is exactly

$$U_\ell + p_\ell V_\ell = -2\pi\mathrm{i}. \tag{5.3}$$

By (1.4) these are precisely the critical equations of (1.5). Thus local uniqueness applies to the stated critical point, not only to a separate system of period equations.

There is also a direct classical check that the filled auxiliary groups have a genuine loxodromic direction. Separate evenness of Φ_ε gives holomorphic functions $\widetilde{\tau}_\ell$ such that

$$V_\ell = \widetilde{\tau}_\ell(\boldsymbol{\chi})\, U_\ell, \qquad \widetilde{\tau}_\ell(0) = \tau_\ell.$$

In fact divide $\partial_{\chi_\ell}\Phi_\varepsilon$ by $\pi\mathrm{i}\chi_\ell$; the apparent singularity is removable. Shrink the neighborhood so $\operatorname{Im}\widetilde{\tau}_\ell > 0$ throughout it. At the solution of (5.1),

$$\begin{aligned} \chi_{\ell,*} &= \frac{r_\ell}{\widetilde{\tau}_\ell(\boldsymbol{\chi}_*) + r_\ell}, \\ \operatorname{Re} V_\ell &= \frac{2\pi r_\ell^2 \operatorname{Im}\widetilde{\tau}_\ell(\boldsymbol{\chi}_*)}{|\widetilde{\tau}_\ell(\boldsymbol{\chi}_*) + r_\ell|^2} > 0. \end{aligned} \tag{5.4}$$

The second formula holds for either sign of the nonzero real r_ℓ. Thus the longitude multiplier has nonunit modulus. The sign is derived from the classical potential and the original positive cusp ratio, without a source noncancellation argument.

6 From a peripheral equation to the original smooth filling

We now apply a metric result, rather than identifying an exponential equation with a filled metric. The following is the precise input [3, Proposition 6.1].

Proposition 6.1 (Near-complete primitive completion, imported)**.** *Let M be an oriented complete finite-volume hyperbolic three-manifold with torus cusps and complete holonomy ρ_0. Fix primitive bases (μ_i, λ_i). There is one neighborhood of ρ_0 with its peripheral fixed flags, chosen before any slopes, with the following property. Let ρ be an actual representation in that neighborhood and U_i, V_i its small similarity logarithms in bounded frames sending the flags to infinity. Suppose both multipliers are one at each cusp to remain unfilled. At each cusp to be filled suppose $a_iU_i + b_iV_i = \pm 2\pi\mathrm{i}$ with $\gcd(a_i, b_i) = 1$. Then ρ descends to the original topological filling and is its smooth complete finite-volume hyperbolic holonomy.*

This statement supplies completion; it does not supply the representation or solve its equations. Its hypotheses hold here for the following concrete reasons. The complete manifold is the actual Y_E^ε of [1, Theorem 1.2]. Its primitive bases are the traced curves (4.1). The rational face maps and original cut relations prove an actual framed representation near that holonomy, in bounded charts. Section 4 proves that every main peripheral group is a translation group. The equations at the auxiliary cusps are (5.3); their slopes are primitive because $(1, p_\ell)$ is a primitive integer vector in the original basis. These facts verify every premise of Proposition 6.1.

For clarity, we spell out the winding step in that proposition and its relation to the present markings. On a compact truncated core, a finite partition of unity in the future hyperboloid model averages the small change of complete developing maps:

$$\begin{aligned} Y_\rho(x) &= \sum_{j,\gamma} \eta_{j,\gamma}(x)\rho(\gamma)\rho_0(\gamma)^{-1}x, \\ E_\rho(x) &= \frac{Y_\rho(x)}{\sqrt{-\langle Y_\rho(x), Y_\rho(x)\rangle}}. \end{aligned} \tag{6.1}$$

Here the $\eta_{j,\gamma}$ are equivariant lifts of a finite partition on evenly covered balls. A positive sum of future timelike vectors is timelike. Reindexing proves equivariance, and on a compact fundamental part E_ρ is C^1-close to the complete identity map. It is therefore a positive local diffeomorphism. This argument assumes no discreteness of the deformed representation.

For one complete cusp use coordinates $(w, s) \in \mathbb{C} \times [s_0, \infty)$ with translation lattice Λ. In the moving flag frame its peripheral maps are $A_\lambda(z) = a_\lambda z + b_\lambda$. Choose a smooth compactly supported partition satisfying $\sum_{\lambda\in\Lambda} \eta(w - \lambda) = 1$, and put

$$\begin{aligned} d_\rho(w) &= \sum_{\lambda\in\Lambda} \eta(w - \lambda) A_\lambda(w - \lambda), \\ F_\rho(w, s) &= \big(d_\rho(w), e^{s+\xi_\rho(w)}\big), \end{aligned} \tag{6.2}$$

where ξ_ρ is the real-linear extension of $\lambda \mapsto \log|a_\lambda|$. Then $d_\rho(w + \lambda) = A_\lambda d_\rho(w)$. Its real Jacobian is positive on a compact fundamental parallelogram, by closeness to the identity, and equivariance propagates this everywhere. The Jacobian of F_ρ is $e^{s+\xi_\rho(w)} \det_{\mathbb{R}} Dd_\rho(w) > 0$. Interpolation with (6.1) on a compact collar remains C^1-close and positive. This constructs an actual nonsingular hyperbolic metric on the original core and its ideal-end-deleted cusps.

At a cusp to be filled, (5.4) in particular gives a multiplier different from one. Commuting affine maps have a common finite fixed point q, in addition to infinity. For any generator with $a \neq 1$ it is $q = b/(1 - a)$. Both complete generator translations are nonzero, while their multipliers tend to one. Thus $|q| \to \infty$ uniformly near completeness, even if one generator has multiplier one. The developed fundamental parallelogram and its fixed generator paths remain in a bounded set and have bounded lengths. They therefore miss q in one sufficiently small neighborhood. Equivariance extends this exclusion to the entire link cover.

On that simply connected cover choose $f = \mathrm{Log}(d_\rho - q)$. The metric $|df|^2$ is nonsingular and descends to a compact torus; it is complete. Its lifted Euclidean developing map covers $\mathbb{C}$, by lifting finite line segments. Since the plane is simply connected, f is a global diffeomorphism and its periods form a rank-two lattice. For a fixed generator path of developed length at most L, inside $|d_\rho| \leq R$,

$$|\Delta f| \leq \frac{L}{|q| - R} \longrightarrow 0. \tag{6.3}$$

Its exponential is the actual affine multiplier. Consequently the increment equals its prescribed small logarithm, with no extra $2\pi\mathrm{i}$ integer. Thus the two actual periods are U, V before the potentially large coefficient p is used. This is why the fixed neighborhood controls all independently long slopes.

Translate q to zero. Periodicity bounds $e^{s_0+\xi_\rho(w)}/|d_\rho(w) - q|$, so the end contains a full epigraph $y > A|d_\rho - q|$. Set

$$r = \frac{y}{|d_\rho - q|}, \qquad R = \operatorname{arsinh}(1/r), \qquad \theta = \operatorname{Im} f, \qquad X = \operatorname{Re} f + \tfrac{1}{2}\log(1 + r^2).$$

Direct substitution gives

$$ds^2 = dR^2 + \cosh^2 R\, dX^2 + \sinh^2 R\, d\theta^2. \tag{6.4}$$

The epigraph contains a whole punctured tube $0 < R < R_0$. A period L translates (X, θ) by $(\operatorname{Re} L, \operatorname{Im} L)$. The original slope $\mu + p\lambda$ has period $-2\pi\mathrm{i}$, exactly one angular turn. Its complementary primitive period V has nonzero real part, either by rank two of the period lattice or directly by (5.4). Adding the axis gives a smooth solid torus, with precisely the original slope as meridian. Its holonomy is the identity: it fixes both q, ∞ and has multiplier one. This proves group descent, rather than merely an exponential relation. Reversing the angular sign reverses its parametrization, not the metric orientation.

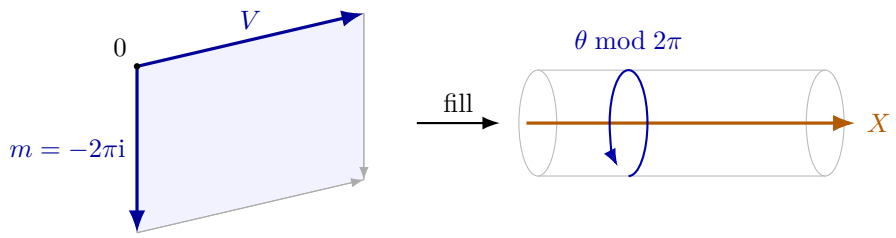


Figure 2: The actual logarithmic link and its smooth filling. The period $m = U + pV = -2\pi\mathrm{i}$ is the primitive meridian of the added solid torus; V is the original disk-longitude period. The old basis is $U = m - pV$. Equation (6.4) turns V into an axial translation and rotation, and m into one full angular turn.

At an unfilled main cusp the multipliers are one. The same compact Euclidean-link argument now gives a complete translation plane and a full horoball quotient in (6.2). The middle region is compact; the added tubes and the remaining horoball ends have finite volume. The metric is complete. Its universal developing map covers simply connected hyperbolic space and hence is an isometry. The filled topology and all its markings are the ones described in (1.2). This completes the metric part of Theorem 1.1.

7 The absolute action and the filled fundamental chain

7.1 The correct polarization

Define $G = 2\pi\mathrm{i}\Phi_\varepsilon$. The exact period formula (1.4) gives

$$\mathrm{d}G = \frac{1}{2}\sum_\ell V_\ell \,\mathrm{d}U_\ell, \qquad \operatorname{Im} G(0) = 2\pi F_0 = \operatorname{Vol}(Y_E^\varepsilon). \tag{7.1}$$

This is the $V\,dU$ polarization. A potential with differential $-\frac{1}{2}\sum U\,dV$ differs by the boundary term $\frac{1}{2}\sum UV$. Its regulator correction cannot be copied unchanged into (7.1).

Let $\mathcal{V}(\rho_\chi)$ be the signed ideal-tetrahedron volume of the retained original fundamental chain from Section 3. The local peripheral differential [3, equation (7.3)] is

$$\mathrm{d}\mathcal{V} = \frac{1}{2}\sum_\ell (\operatorname{Re} V_\ell \,\mathrm{d}\operatorname{Im} U_\ell - \operatorname{Re} U_\ell \,\mathrm{d}\operatorname{Im} V_\ell). \tag{7.2}$$

Its hypotheses are present here: the oriented ideal chain is positive and nondegenerate near the complete point; actual face maps cancel the internal boundaries; edge relations keep their logarithmic sums fixed; and the primitive peripheral basis has the orientation $\operatorname{Im}(\lambda/\mu) > 0$ at completeness. All main logs vanish because their peripheral actions are translations, so they contribute zero. The factor $1/2$ in (7.2) is for similarity logarithms; replacing them by eigenvalue logarithms without adjusting the factor would change the formula.

For reference, that local identity follows by differentiating the Bloch–Wigner volume on each oriented tetrahedron and cancelling paired normal-face flows. The remaining cusp coefficient rows have intersection pairing -2 on the primitive meridian/longitude basis; the inverse tangent pairing is $-1/2$. In [3, Section 7.1] the signs are computed with the positively ordered model $(0, \infty, 1, z)$ in standard upper-half-space orientation. The original complete cusp orientation (1.3) is precisely that convention. No complete character-variety uniqueness is an input to this local calculation.

Taking the imaginary part of (7.1) shows that the differential of $\operatorname{Im} G - \frac{1}{2}\sum \operatorname{Re} U_\ell \operatorname{Im} V_\ell$ is exactly (7.2). At completeness its value is the actual metric volume, by (7.1); hence the constant on this connected neighborhood is fixed:

$$\mathcal{V}(\rho_{\boldsymbol{\chi}}) = 2\pi \operatorname{Re} \Phi_\varepsilon(\boldsymbol{\chi}) - \frac{1}{2}\sum_\ell \operatorname{Re} U_\ell \ \operatorname{Im} V_\ell. \tag{7.3}$$

This is an absolute equality on the original marked chain.

7.2 The critical boundary term

At a real integer filling tuple, write $\chi_\ell = u_\ell + \mathrm{i}v_\ell$. The critical equation gives $V_\ell = 2\pi\mathrm{i}(\chi_\ell - 1)/p_\ell$. Direct substitution yields

$$\begin{aligned} -\tfrac{1}{2} \operatorname{Re} U_\ell \ \operatorname{Im} V_\ell &= -\frac{2\pi^2}{p_\ell} v_\ell (u_\ell - 1) \\ &= 2\pi \operatorname{Re}\left(\frac{\pi\mathrm{i}}{2p_\ell}(\chi_\ell - 1)^2\right). \end{aligned} \tag{7.4}$$

Thus (7.3) equals $2\pi \operatorname{Re} S_{\varepsilon,\mathbf{p}}(\boldsymbol{\chi}_*)$. It remains to identify the same ideal-chain volume with the complete filled metric, rather than infer that conclusion from a differential identity.

7.3 Keeping the same boundary triangulation

The comparison in [3, Section 7.3] applies to the primitive completion proved above. We describe its chain content to fix the absolute interpretation. At a filled cusp conjugate the two fixed points to $0, \infty$. On each triangle of the retained boundary triangulation the conical ideal development has form $(a(w), Yb(w))$, with $b > 0$. The functions and their derivatives are bounded on a finite fundamental set. An equivariant axis map is $(0, Ye^{\eta(w)})$, where η has the real peripheral logarithmic character. On that same triangulation use

$$H_s(w, Y) = \big(sa(w), Y \exp\big((1-s)\eta(w) + s \log b(w)\big)\big), \qquad 0 \le s \le 1. \tag{7.5}$$

It is face-compatible and equivariant: horizontal coordinates transform by the complex multiplier and both heights by its modulus. The height is comparable to Y, horizontal derivatives are bounded, and height derivatives are $O(Y)$. The hyperbolic volume form $y^{-3}\, dx_1 \wedge dx_2 \wedge dy$ bounds this boundary homotopy's volume by $O(Y^{-2})$.

After the primitive slope has been killed, extend the same axis-valued triangulated boundary through the solid torus. The remaining cyclic peripheral group acts along the axis. All added vertices lie on that axis, so every added tetrahedron has zero signed volume. Unfilled main cusp comparisons have the same vanishing boundary estimate. Stokes on finite truncations, followed by $Y \to \infty$, therefore identifies the original chain with the filled equivariant fundamental class.

Compare it with the developing map of the complete metric from Section 6. They represent this same class and have the same remaining cusp flags. A common subdivision, straightening and the same vanishing boundary estimate give

$$\mathcal{V}(\rho_{\boldsymbol{\chi}_*}) = \operatorname{Vol}(Y_E^\varepsilon(\mathbf{p})).$$

Using an unrelated new triangulation with axis vertices would not prove this equality; it is the retained boundary homotopy (7.5) that supplies the comparison. The usual metric volume is unambiguous also by the volume-independence statement [3, Lemma 7.1]. Together with (7.4) this proves (1.6) and Theorem 1.1.

8 The analytic interface and finite checks

The classical potential has been defined from the cell functions (2.1), (2.2) and the exact meridian shear. An application to a finite source must separately prove that its selected limiting leaf action is this f_ε, including the parameter-dependent constant. Matching only its x derivative or its complete Hessian is insufficient. For example the exact classical exponential relation $R^{-1} - 1 = e^{-\partial_x f_\varepsilon}$ fixes a face shape, but does not by itself identify a finite coefficient's normalization. One must also compare the endpoint-normalized leaf primitive and then eliminate the same internal wires. Control of complementary terms and a nonzero whole filling asymptotic are further analytic assertions. None is a premise or conclusion of Theorem 1.1.

The construction-specific imported inputs are the complete stationary point of [2, Proposition 5.3] and the complete metric, actual cut and primitive bases of [1, Theorem 1.2 and Sections 4–7]. The full complex continuation uses the latter paper's rational face identities, not a claim that a nonplanar deformed half ball is convex. The metric completion and peripheral-volume facts imported from [3] have their actual hypotheses verified above. Standard holomorphic implicit-function theory is used only at the two explicitly invertible Jacobians. The usual Dehn-filling and peripheral-volume framework provides historical context [4]; the original words and the absolute normalization are retained in the calculation here.

The deterministic accompanying program `check.py` verifies exact rational flag and multiplier identities, the local Hessian coefficient, the real signed formula (5.4), and the boundary cancellation (7.4). Its negative controls detect the missing factor two between eigenvalue and similarity logs and an incorrect shear sign. The record includes hashes of the actual source inputs. These checks establish finite algebraic equalities only. They do not prove the implicit-function theorem, the original topological identification, metric completeness, or the fundamental-chain comparison.

References

[1] *Pyramids, prisms, and complete geometry of Conway closures*, companion paper, Theorem 1.2 and Sections 4–7. Companion PDF.

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[4] W. D. Neumann and D. Zagier, Volumes of hyperbolic three-manifolds, *Topology* **24** (1985), 307–332. doi:10.1016/0040-9383(85)90004-7.